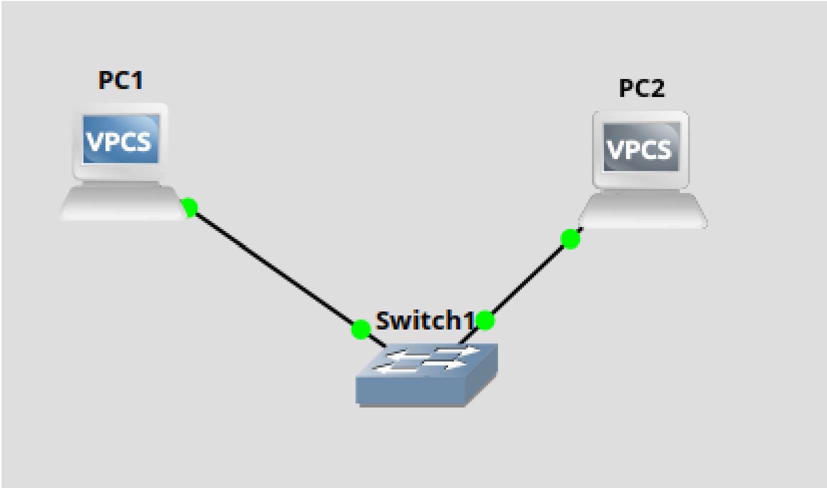


Рисунок1 – Схема сети

Для того чтобы задать ip адрес PC испотзуем команды

ip 192.168.1.10/24 192.168.1.1

ip 192.168.1.20/24 192.168.1.1

для проверки введем show ip

```
PC1> ip 192.168.1.10/24 192.168.1.1
Checking for duplicate address...
PC1 : 192.168.1.10 255.255.255.0 gateway 192.168.1.1

PC1>
```

Рисунок 2 – Настройка сети PC1

```
PC1> show ip

NAME        : PC1[1]
IP/MASK     : 192.168.1.10/24
GATEWAY     : 192.168.1.1
DNS         :
MAC         : 00:50:79:66:68:00
LPORT       : 10006
RHOST:PORT  : 127.0.0.1:10007
MTU         : 1500

PC1>
```

Рисунок 3 – Вывод настройки сети PC1

```
PC2> ip 192.168.1.20/24 192.168.1.1
Checking for duplicate address...
PC2 : 192.168.1.20 255.255.255.0 gateway 192.168.1.1
```

Рисунок 4 – Настройка сети PC2

```
PC2> show ip

NAME        : PC2[1]
IP/MASK     : 192.168.1.20/24
GATEWAY     : 192.168.1.1
DNS         :
MAC         : 00:50:79:66:68:01
LPORT       : 10004
RHOST:PORT  : 127.0.0.1:10005
MTU         : 1500

PC2>
```

Рисунок 5 – Вывод настройки сети PC2

Выполним команду ping для проверки доступности устройсва

```
PC1> ping 192.168.1.20

84 bytes from 192.168.1.20 icmp_seq=1 ttl=64 time=1.341 ms
84 bytes from 192.168.1.20 icmp_seq=2 ttl=64 time=2.157 ms
84 bytes from 192.168.1.20 icmp_seq=3 ttl=64 time=2.998 ms
84 bytes from 192.168.1.20 icmp_seq=4 ttl=64 time=2.703 ms
84 bytes from 192.168.1.20 icmp_seq=5 ttl=64 time=2.715 ms

PC1>
```

Рисунок 6 – Проверка доступности PC2 с PC1

Далее показаны ARP пакеты request и reply

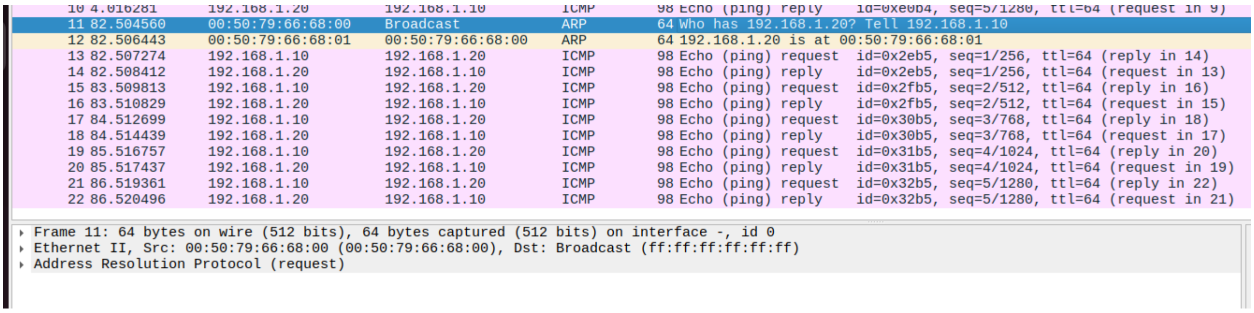


Рисунок 7 – ARP пакет request между switch и PC1

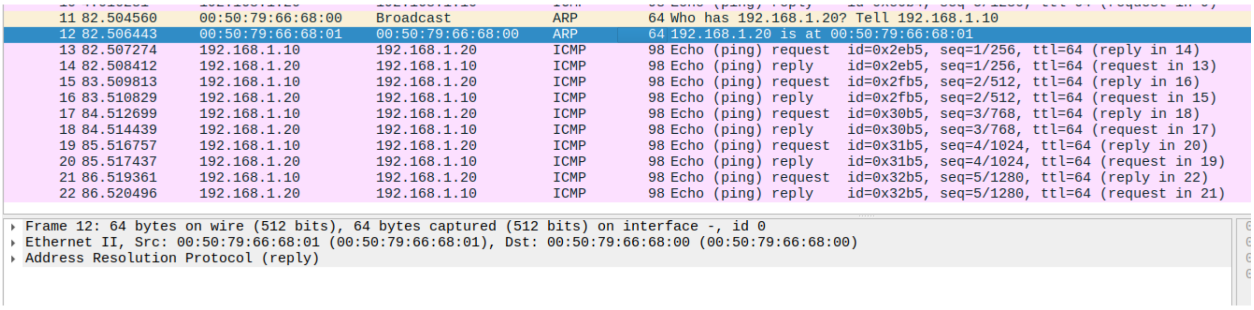


Рисунок 8 – ARP пакет reply между switch и PC1

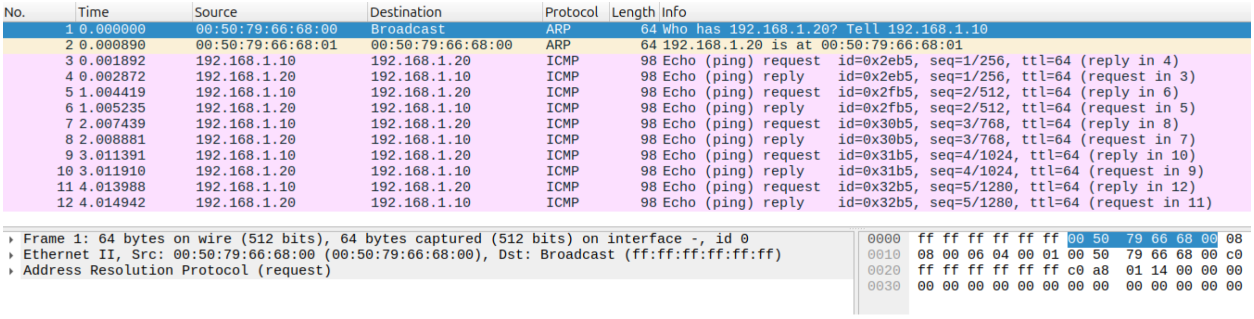


Рисунок 9 – ARP пакет request между switch и PC2

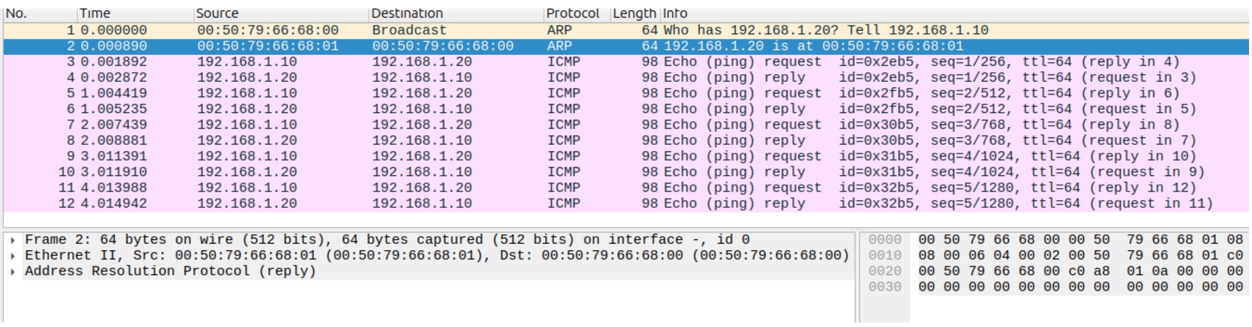


Рисунок 10 – ARP пакет reply между switch и PC2

Маршрутизатор

Для настройки маршрутизатора перейдем в режим конфигурации

Configure terminal

```
R1#enable
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#
```

Рисунок 11 – Переход в режим конфигурации на маршрутизаторы

Для настройки первого интерфейса введем

Interface fastEthernet0/0

Ip address 192.168.1.1 255.255.255.0

No shutdown

Exit

```
R1(config)#interface fastEthernet0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#
*Mar  1 00:05:29.647: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state t
o up
*Mar  1 00:05:30.647: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/0, changed state to up
R1(config)#
```

Рисунок 12 – Настройка интерфейса fastEthetnet0/0

Для второго интерфейса такие же действия

Interface fastEthernet0/1

Ip address 10.0.0.1 255.255.255.0

No shutdown

exit

```
R1(config)#interface fastEthernet0/1
R1(config-if)#ip address 10.0.0.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#
*Mar  1 00:07:17.411: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state t
o up
*Mar  1 00:07:18.411: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/1, changed state to up
R1(config)#
```

Рисунок 13 – Настройка интерфейса fastEthetnet0/1

Для сохранения конфигурации введем

Write memory

```
R1(config)#end
R1#
*Mar  1 00:08:03.571: %SYS-5-CONFIG_I: Configured from console by console
R1#write memory
Building configuration...
[OK]
R1#
Copyright (c) 2006-2025 GNS3 Technologies.
```

Рисунок 14 – Завершение настройки

Настроим ip адреса на PC

Ip 10.0.0.10/24 10.0.0.1

Ip 192.168.1.10/24 192.168.1.1

Проверка доступности ping 10.0.0.10 с 192.168.1.10

```
PC2> ip 10.0.0.10/24 10.0.0.1
Checking for duplicate address...
PC2 : 10.0.0.10 255.255.255.0 gateway 10.0.0.1

PC2>
```

Рисунок 15 – Настройка сети PC2

```
PC1> ip 192.168.1.10/24 192.168.1.1
Checking for duplicate address...
PC1 : 192.168.1.10 255.255.255.0 gateway 192.168.1.1

PC1> ping 10.0.0.10

10.0.0.10 icmp_seq=1 timeout
10.0.0.10 icmp_seq=2 timeout
84 bytes from 10.0.0.10 icmp_seq=3 ttl=63 time=16.149 ms
84 bytes from 10.0.0.10 icmp_seq=4 ttl=63 time=13.806 ms
84 bytes from 10.0.0.10 icmp_seq=5 ttl=63 time=21.030 ms
```

Рисунок 16 – Настройка сети PC1 и проверка доступности PC2

ARP пакеты

arp or icmp

No.	Time	Source	Destination	Protocol	Length	Info
15	111.527922	00:50:79:66:68:01	Broadcast	ARP	64	Who has 192.168.1.1? Tell 192.168.1.10
16	111.532357	c4:01:1c:ad:00:00	00:50:79:66:68:01	ARP	60	192.168.1.1 is at c4:01:1c:ad:00:00
17	111.534204	192.168.1.10	10.0.0.10	ICMP	98	Echo (ping) request id=0xf7bd, seq=1/256, ttl=64 (reply in 18)
18	111.564503	10.0.0.10	192.168.1.10	ICMP	98	Echo (ping) reply id=0xf7bd, seq=1/256, ttl=63 (request in 17)
19	112.567155	192.168.1.10	10.0.0.10	ICMP	98	Echo (ping) request id=0xf8bd, seq=2/512, ttl=64 (reply in 20)
20	112.584041	10.0.0.10	192.168.1.10	ICMP	98	Echo (ping) reply id=0xf8bd, seq=2/512, ttl=63 (request in 19)
21	113.585571	192.168.1.10	10.0.0.10	ICMP	98	Echo (ping) request id=0xf9bd, seq=3/768, ttl=64 (reply in 22)
22	113.600333	10.0.0.10	192.168.1.10	ICMP	98	Echo (ping) reply id=0xf9bd, seq=3/768, ttl=63 (request in 21)
23	114.602433	192.168.1.10	10.0.0.10	ICMP	98	Echo (ping) request id=0xfabd, seq=4/1024, ttl=64 (reply in 24)
24	114.616480	10.0.0.10	192.168.1.10	ICMP	98	Echo (ping) reply id=0xfabd, seq=4/1024, ttl=63 (request in 23)
25	115.618673	192.168.1.10	10.0.0.10	ICMP	98	Echo (ping) request id=0xfbbd, seq=5/1280, ttl=64 (reply in 26)
26	115.633917	10.0.0.10	192.168.1.10	ICMP	98	Echo (ping) reply id=0xfbbd, seq=5/1280, ttl=63 (request in 25)

```
Frame 15: 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, id 0
Ethernet II, Src: 00:50:79:66:68:01 (00:50:79:66:68:01), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
Address Resolution Protocol (request)
```

Рисунок 17 – ARP пакет request между маршрутизатором и PC1

arp or icmp

No.	Time	Source	Destination	Protocol	Length	Info
15	111.527922	00:50:79:66:68:01	Broadcast	ARP	64	Who has 192.168.1.1? Tell 192.168.1.10
16	111.532357	c4:01:1c:ad:00:00	00:50:79:66:68:01	ARP	60	192.168.1.1 is at c4:01:1c:ad:00:00
17	111.534204	192.168.1.10	10.0.0.10	ICMP	98	Echo (ping) request id=0xf7bd, seq=1/256, ttl=64 (reply in 18)
18	111.564503	10.0.0.10	192.168.1.10	ICMP	98	Echo (ping) reply id=0xf7bd, seq=1/256, ttl=63 (request in 17)
19	112.567155	192.168.1.10	10.0.0.10	ICMP	98	Echo (ping) request id=0xf8bd, seq=2/512, ttl=64 (reply in 20)
20	112.584041	10.0.0.10	192.168.1.10	ICMP	98	Echo (ping) reply id=0xf8bd, seq=2/512, ttl=63 (request in 19)
21	113.585571	192.168.1.10	10.0.0.10	ICMP	98	Echo (ping) request id=0xf9bd, seq=3/768, ttl=64 (reply in 22)
22	113.600333	10.0.0.10	192.168.1.10	ICMP	98	Echo (ping) reply id=0xf9bd, seq=3/768, ttl=63 (request in 21)
23	114.602433	192.168.1.10	10.0.0.10	ICMP	98	Echo (ping) request id=0xfabd, seq=4/1024, ttl=64 (reply in 24)
24	114.616480	10.0.0.10	192.168.1.10	ICMP	98	Echo (ping) reply id=0xfabd, seq=4/1024, ttl=63 (request in 23)
25	115.618673	192.168.1.10	10.0.0.10	ICMP	98	Echo (ping) request id=0xfbbd, seq=5/1280, ttl=64 (reply in 26)
26	115.633917	10.0.0.10	192.168.1.10	ICMP	98	Echo (ping) reply id=0xfbbd, seq=5/1280, ttl=63 (request in 25)

```
Frame 16: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0
Ethernet II, Src: c4:01:1c:ad:00:00 (c4:01:1c:ad:00:00), Dst: 00:50:79:66:68:01 (00:50:79:66:68:01)
Address Resolution Protocol (reply)
```

Рисунок 18 – ARP пакет reply между маршрутизатором и PC1

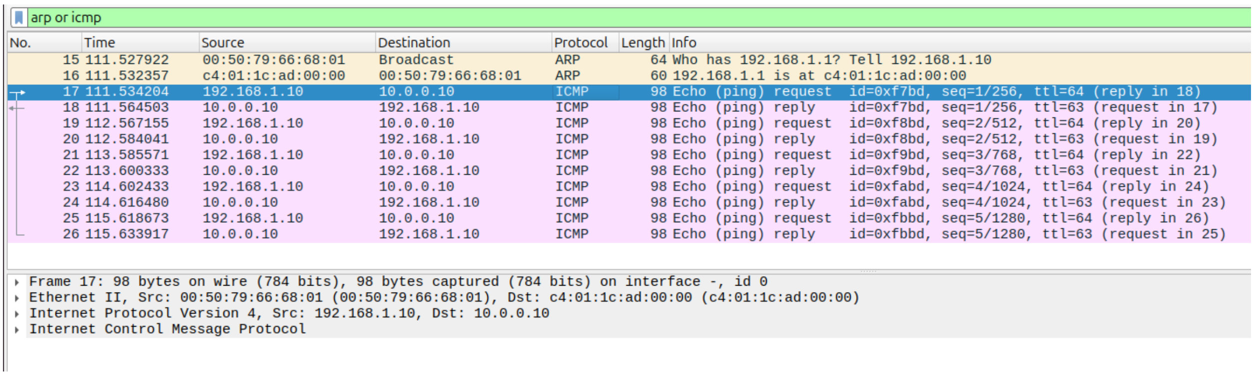


Рисунок 19 – ICMP пакет request между маршрутизатором и PC1

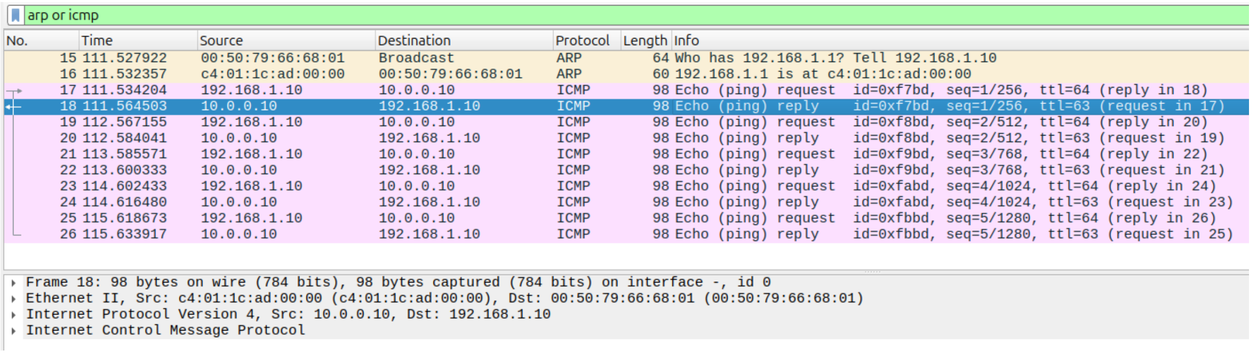


Рисунок 20 – ICMP пакет reply между маршрутизатором и PC1

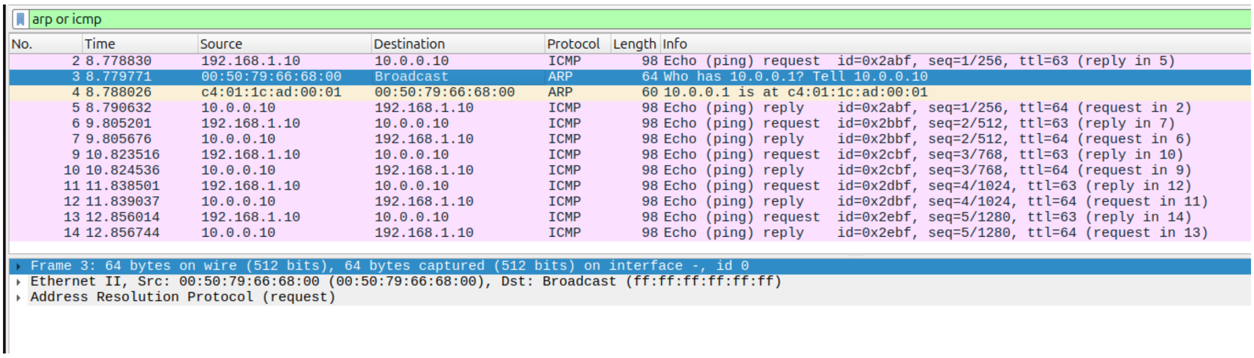


Рисунок 21 – ARP пакет request между маршрутизатором и PC2

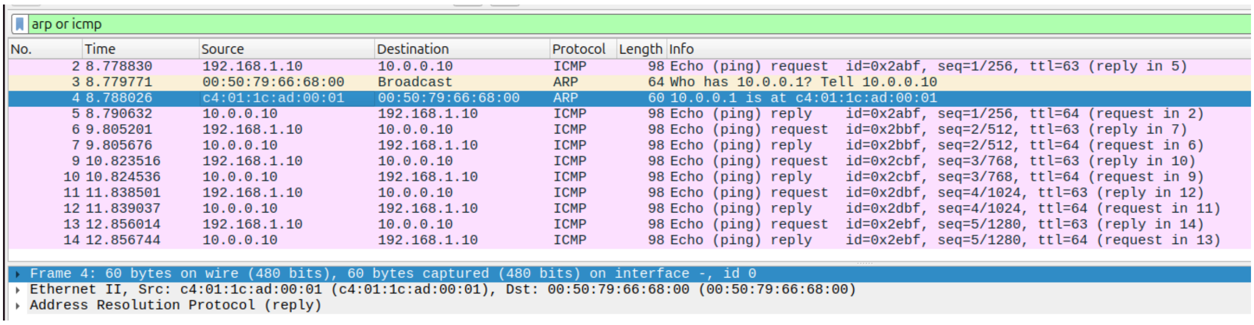
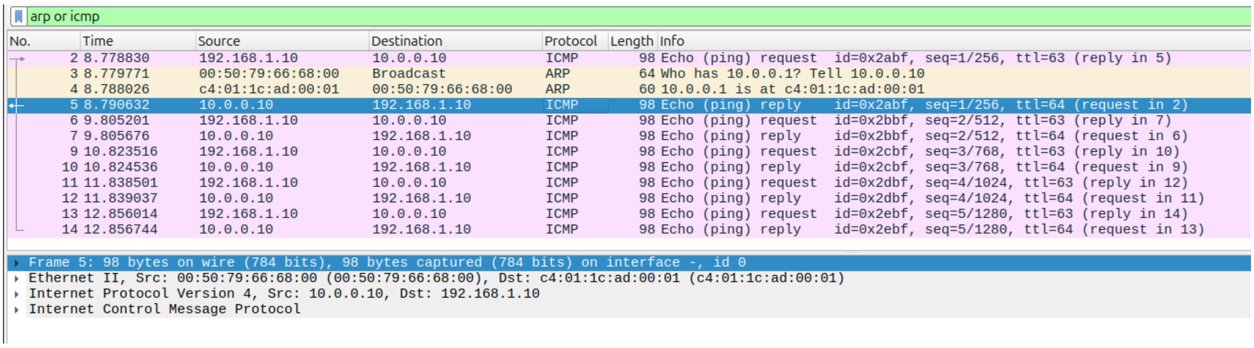


Рисунок 22 – ARP пакет reply между маршрутизатором и PC2

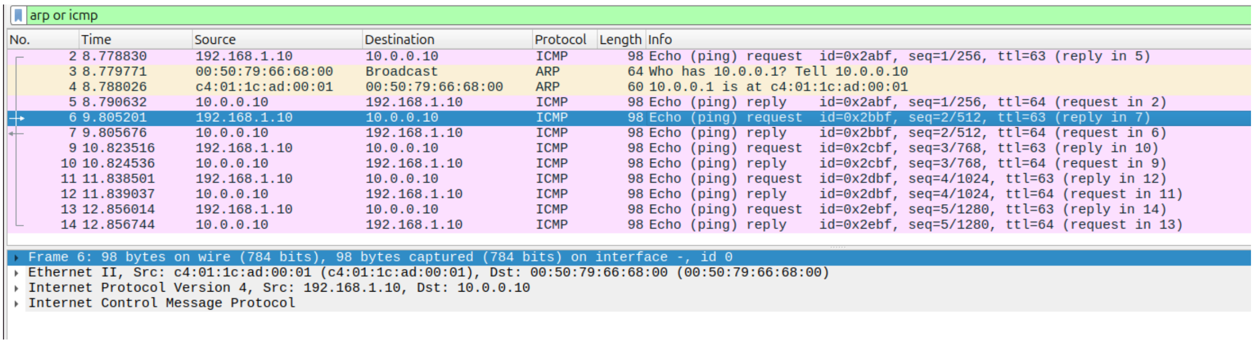


arp or icmp

No.	Time	Source	Destination	Protocol	Length	Info
2	8.778830	192.168.1.10	10.0.0.10	ICMP	98	Echo (ping) request id=0x2abf, seq=1/256, ttl=63 (reply in 5)
3	8.779771	00:50:79:66:68:00	Broadcast	ARP	64	Who has 10.0.0.1? Tell 10.0.0.10
4	8.788026	c4:01:1c:ad:00:01	00:50:79:66:68:00	ARP	60	10.0.0.1 is at c4:01:1c:ad:00:01
5	8.790632	10.0.0.10	192.168.1.10	ICMP	98	Echo (ping) reply id=0x2abf, seq=1/256, ttl=64 (request in 2)
6	9.805201	192.168.1.10	10.0.0.10	ICMP	98	Echo (ping) request id=0x2bbf, seq=2/512, ttl=63 (reply in 7)
7	9.805676	10.0.0.10	192.168.1.10	ICMP	98	Echo (ping) reply id=0x2bbf, seq=2/512, ttl=64 (request in 6)
9	10.823516	192.168.1.10	10.0.0.10	ICMP	98	Echo (ping) request id=0x2cbf, seq=3/768, ttl=63 (reply in 10)
10	10.824536	10.0.0.10	192.168.1.10	ICMP	98	Echo (ping) reply id=0x2cbf, seq=3/768, ttl=64 (request in 9)
11	11.838501	192.168.1.10	10.0.0.10	ICMP	98	Echo (ping) request id=0x2dbf, seq=4/1024, ttl=63 (reply in 12)
12	11.839037	10.0.0.10	192.168.1.10	ICMP	98	Echo (ping) reply id=0x2dbf, seq=4/1024, ttl=64 (request in 11)
13	12.856014	192.168.1.10	10.0.0.10	ICMP	98	Echo (ping) request id=0x2ebf, seq=5/1280, ttl=63 (reply in 14)
14	12.856744	10.0.0.10	192.168.1.10	ICMP	98	Echo (ping) reply id=0x2ebf, seq=5/1280, ttl=64 (request in 13)

Frame 5: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0
Ethernet II, Src: 00:50:79:66:68:00 (00:50:79:66:68:00), Dst: c4:01:1c:ad:00:01 (c4:01:1c:ad:00:01)
Internet Protocol Version 4, Src: 10.0.0.10, Dst: 192.168.1.10
Internet Control Message Protocol

Рисунок 23 – ICMP пакет request между маршрутизатором и PC2



arp or icmp

No.	Time	Source	Destination	Protocol	Length	Info
2	8.778830	192.168.1.10	10.0.0.10	ICMP	98	Echo (ping) request id=0x2abf, seq=1/256, ttl=63 (reply in 5)
3	8.779771	00:50:79:66:68:00	Broadcast	ARP	64	Who has 10.0.0.1? Tell 10.0.0.10
4	8.788026	c4:01:1c:ad:00:01	00:50:79:66:68:00	ARP	60	10.0.0.1 is at c4:01:1c:ad:00:01
5	8.790632	10.0.0.10	192.168.1.10	ICMP	98	Echo (ping) reply id=0x2abf, seq=1/256, ttl=64 (request in 2)
6	9.805201	192.168.1.10	10.0.0.10	ICMP	98	Echo (ping) request id=0x2bbf, seq=2/512, ttl=63 (reply in 7)
7	9.805676	10.0.0.10	192.168.1.10	ICMP	98	Echo (ping) reply id=0x2bbf, seq=2/512, ttl=64 (request in 6)
9	10.823516	192.168.1.10	10.0.0.10	ICMP	98	Echo (ping) request id=0x2cbf, seq=3/768, ttl=63 (reply in 10)
10	10.824536	10.0.0.10	192.168.1.10	ICMP	98	Echo (ping) reply id=0x2cbf, seq=3/768, ttl=64 (request in 9)
11	11.838501	192.168.1.10	10.0.0.10	ICMP	98	Echo (ping) request id=0x2dbf, seq=4/1024, ttl=63 (reply in 12)
12	11.839037	10.0.0.10	192.168.1.10	ICMP	98	Echo (ping) reply id=0x2dbf, seq=4/1024, ttl=64 (request in 11)
13	12.856014	192.168.1.10	10.0.0.10	ICMP	98	Echo (ping) request id=0x2ebf, seq=5/1280, ttl=63 (reply in 14)
14	12.856744	10.0.0.10	192.168.1.10	ICMP	98	Echo (ping) reply id=0x2ebf, seq=5/1280, ttl=64 (request in 13)

Frame 6: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0
Ethernet II, Src: c4:01:1c:ad:00:01 (c4:01:1c:ad:00:01), Dst: 00:50:79:66:68:00 (00:50:79:66:68:00)
Internet Protocol Version 4, Src: 192.168.1.10, Dst: 10.0.0.10
Internet Control Message Protocol

Рисунок 24 – ICMP пакет reply между маршрутизатором и PC2

В ходе работы с помощью Wireshark был перехвачен и проанализирован трафик протоколов ARP и ICMP между PC1, PC2 и маршрутизатором.

ARP-пакеты:

Request: Широковещательные запросы (MAC ff:ff:ff:ff:ff:ff) для определения MAC-адреса по IP (например, PC1 → 192.168.1.1).

Reply: Уникастальные ответы с MAC-адресом целевого устройства.

Вывод: ARP корректно разрешает адреса в пределах каждой подсети.

ICMP-пакеты:

Echo Request (Type 8): Запросы ping между узлами разных подсетей (PC1 → PC2).

Echo Reply (Type 0): Ответы, подтверждающие доступность.

Вывод: Маршрутизатор успешно перенаправляет ICMP-трафик между сетями 192.168.1.0/24 и 10.0.0.0/24.